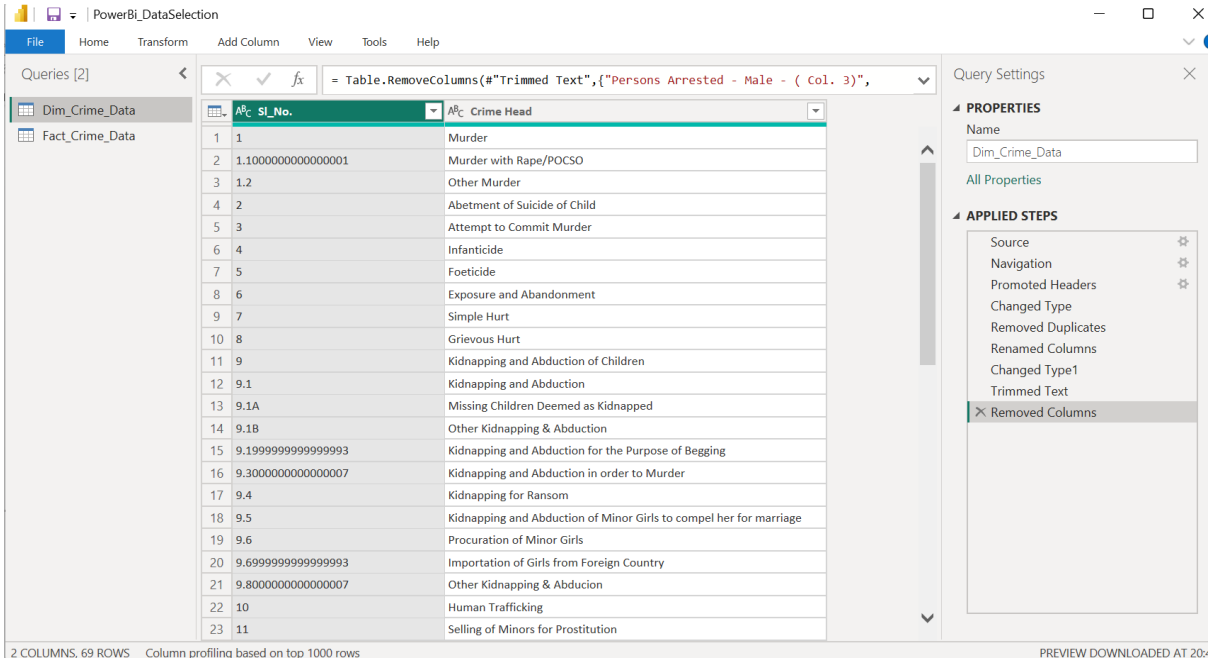
TASK 3 – DATA MODELING

- **Identify Fact table(s) and Dimension table(s).**
- **Build relationships (one-to-many / many-to-one).**
- **Check cardinality and cross-filter direction.**
- ✅ **Task: Create a clear and correct data model.**

Step1: Identify Fact Tables and Dimension Tables

a)Dimension Table



	Sl_No.	Crime Head
1	1	Murder
2	1.1000000000000001	Murder with Rape/POCSO
3	1.2	Other Murder
4	2	Abetment of Suicide of Child
5	3	Attempt to Commit Murder
6	4	Infanticide
7	5	Foeticide
8	6	Exposure and Abandonment
9	7	Simple Hurt
10	8	Grievous Hurt
11	9	Kidnapping and Abduction of Children
12	9.1	Kidnapping and Abduction
13	9.1A	Missing Children Deemed as Kidnapped
14	9.1B	Other Kidnapping & Abduction
15	9.1999999999999993	Kidnapping and Abduction for the Purpose of Begging
16	9.3000000000000007	Kidnapping and Abduction in order to Murder
17	9.4	Kidnapping for Ransom
18	9.5	Kidnapping and Abduction of Minor Girls to compel her for marriage
19	9.6	Procuration of Minor Girls
20	9.6999999999999993	Importation of Girls from Foreign Country
21	9.8000000000000007	Other Kidnapping & Abducion
22	10	Human Trafficking
23	11	Selling of Minors for Prostitution

Dimension Table – Dim_Crime_Data containing descriptive attributes of crime categories.

b) Fact Table

Queries [2]

Dim_Crime_Data

Fact_Crime_Data

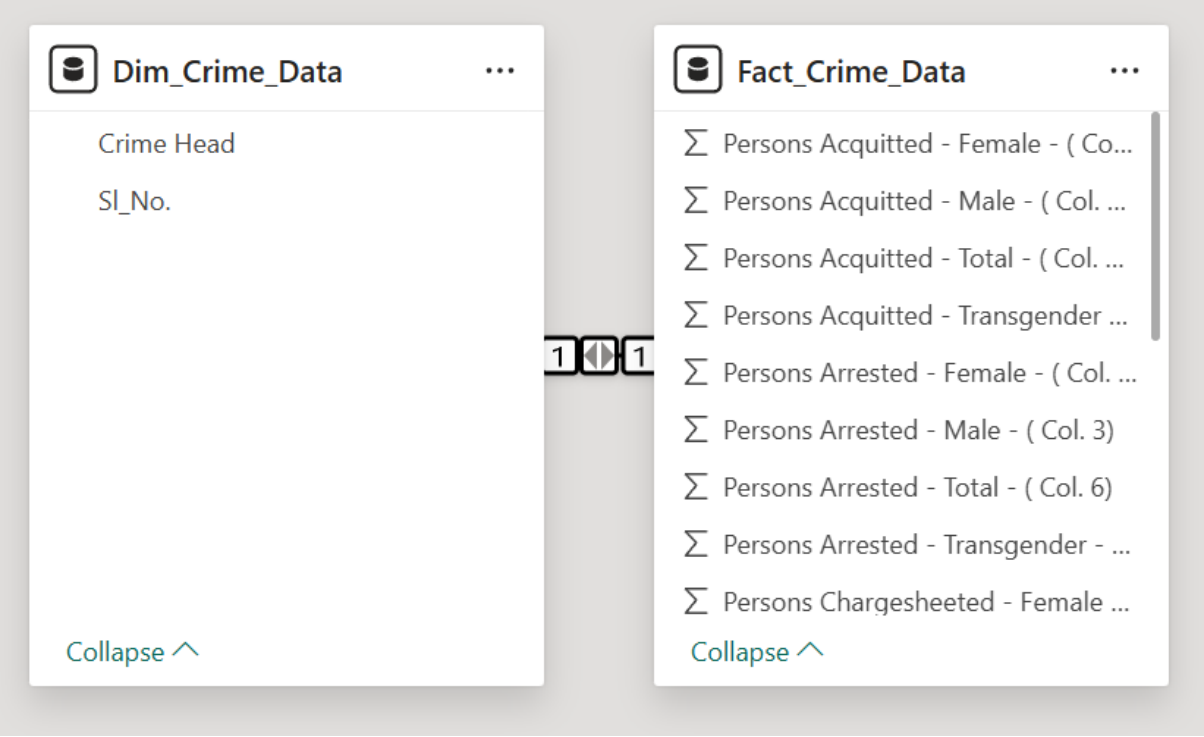
```
= Table.RemoveColumns(#"Trimmed Text",{"Crime Head"})
```

	Sl_No.	Persons Arrested - Male - (Col. 3)	Persons Arrested - Female - (Col. 4)
1	1	1793	
2	1.1000000000000001	151	
3	1.2	1642	
4	2	446	
5	3	553	
6	4	42	
7	5	64	
8	6	135	
9	7	1880	
10	8	500	
11	9	31678	
12	9.1	16156	
13	9.1A	8464	
14	9.1B	7692	
15	9.1999999999999993	77	
16	9.3000000000000007	70	
17	9.4	171	
18	9.5	11161	
19	9.6	1511	
20	9.6999999999999993	5	
21	9.8000000000000007	2527	
22	10	608	
23			

21 COLUMNS, 69 ROWS Column profiling based on top 1000 rows

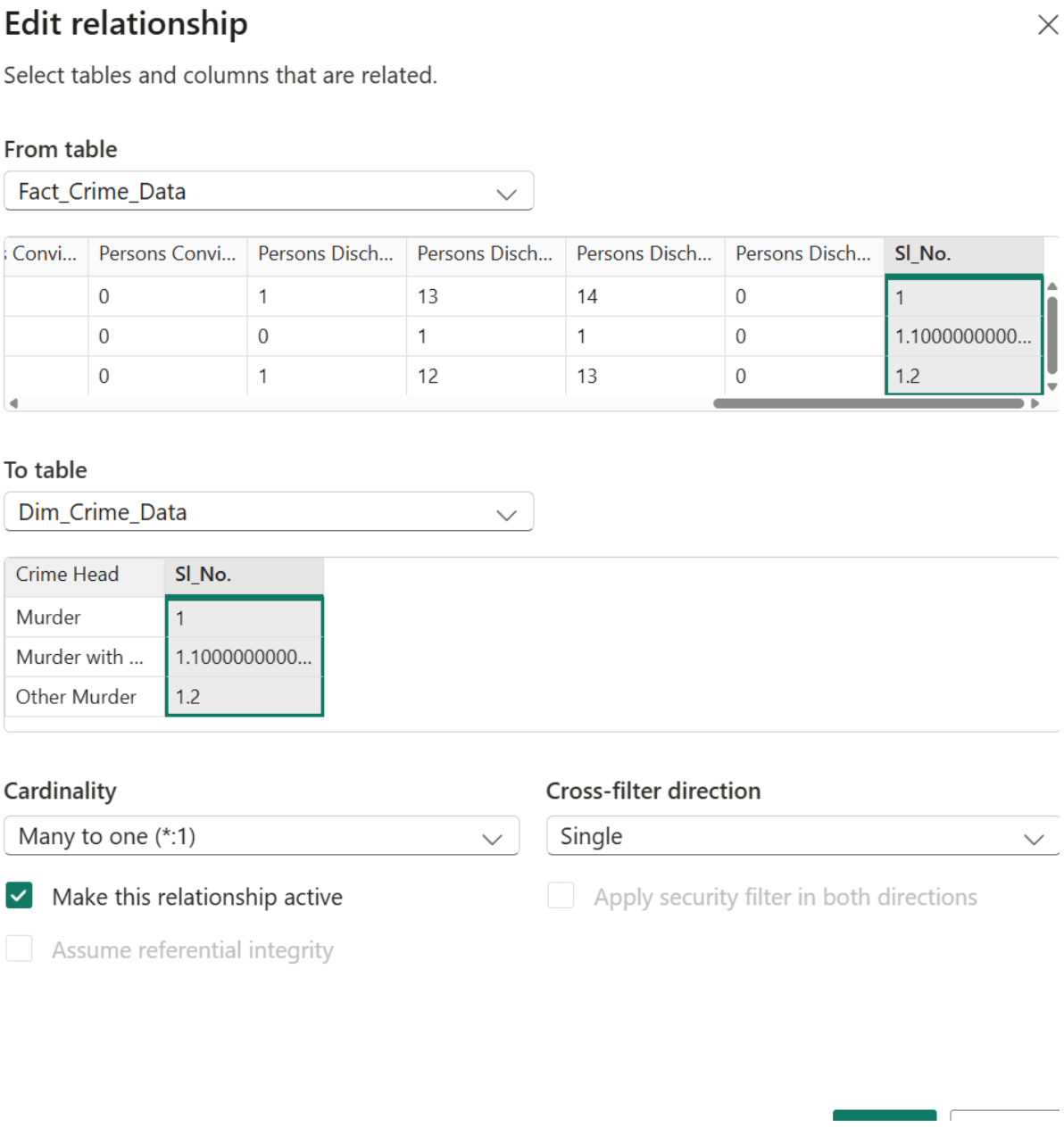
Fact Table – Fact_Crime_Data containing numerical measures related to crime statistics.

Step2: Build relationships (one-to-many / many-to-one).



Data Model showing relationship between Dimension and Fact tables.

Step3: Check cardinality and cross-filter direction.



A relationship was created between Dim_Crime_Data and Fact_Crime_Data using the Sl_No column.
The relationship has One-to-Many (1:*) cardinality, where the Dimension table contains unique values and the Fact table contains multiple related records.
The Cross-filter direction is set to Single, allowing filters to flow from the Dimension table to the Fact table, ensuring efficient data modeling.